

MUNICH INTL, Germany

WMO: 108660

Lat: 48.3478N

Lon: 11.8133E

Elev: 453

StdP: 96.00

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-9.8	-15.0	1.1	-11.9	-11.9	1.4	-8.7	15.2	8.6	13.3	6.9	2.1	80	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	30.1	19.3	28.1	18.7	26.2	18.1	20.3	27.2	19.6	26.2	18.8	24.9	3.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.8	22.5	17.2	13.0	21.6	16.8	12.7	21.3	60.6	27.6	57.8	26.0	55.4	24.9	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.5	8.1	DB	-16.8	32.7	3.6	1.9	-19.4	34.1	-21.5	35.2	-23.6	36.3	-26.2	37.7	(4)
(5)			WB	-17.0	22.2	3.6	1.1	-19.6	23.0	-21.7	23.6	-23.7	24.2	-26.4	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-0.4	0.8	4.8	9.1	13.6	17.0	18.8	18.4	13.8	9.5	4.1	0.8	(6)
(7)		DBStd	7.83	4.85	4.91	4.09	3.84	3.48	3.54	2.98	3.02	3.21	3.70	3.94	4.37	(7)
(8)		HDD10.0	1339	324	257	164	61	8	1	0	0	5	55	179	286	(8)
(9)		HDD18.3	3436	582	489	419	277	152	66	31	37	140	274	426	543	(9)
(10)		CDD10.0	1062	1	1	4	34	119	212	272	260	118	39	3	0	(10)
(11)		CDD18.3	119	0	0	0	0	4	27	45	39	3	0	0	0	(11)
(12)		CDH23.3	1337	0	0	0	7	76	295	479	428	50	2	0	0	(12)
(13)		CDH26.7	350	0	0	0	1	8	69	141	128	4	0	0	0	(13)
(14)	Wind	WSAvg	3.3	3.8	3.8	4.2	3.5	3.3	2.9	3.0	2.7	2.8	2.9	3.3	3.6	(14)
(15)	Precipitation	PrecAvg	839	50	43	55	51	90	103	106	95	70	59	59	57	(15)
(16)		PrecMax	1022	143	89	134	110	150	172	218	167	138	131	120	100	(16)
(17)		PrecMin	561	2	10	16	8	12	45	24	38	23	10	0	4	(17)
(18)		PrecStd	95	29	23	31	28	36	38	42	37	29	29	27	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	15.0	19.0	24.1	27.9	31.2	32.2	33.0	27.1	22.8	17.0	12.8	(19)
(20)			MCWB	8.8	8.8	10.9	13.7	17.5	20.0	20.0	19.4	18.0	15.6	12.2	8.9	(20)
(21)		2%	DB	10.9	11.9	16.2	21.9	25.8	28.9	30.1	30.0	24.9	20.0	14.1	10.2	(21)
(22)			MCWB	7.7	7.6	9.6	12.9	16.7	19.6	19.3	19.3	17.5	14.9	10.7	7.7	(22)
(23)		5%	DB	8.2	10.0	13.9	19.2	23.1	26.9	28.0	27.8	22.8	17.9	11.9	8.8	(23)
(24)			MCWB	6.0	6.6	8.9	11.8	15.4	18.5	18.7	18.7	16.7	13.7	9.3	6.7	(24)
(25)		10%	DB	6.1	7.9	11.8	17.1	21.0	24.9	26.0	25.8	20.2	15.8	10.0	6.9	(25)
(26)			MCWB	4.6	5.3	7.9	10.6	14.2	17.7	18.3	18.1	15.5	12.9	8.2	5.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	9.4	11.8	15.0	18.6	21.7	21.5	21.4	19.2	16.4	12.6	9.6	(27)
(28)			MCDWB	12.2	13.5	16.7	22.7	25.3	28.5	29.1	28.4	24.6	20.9	16.0	11.8	(28)
(29)		2%	WB	7.8	8.0	10.3	13.5	17.3	20.1	20.2	20.1	18.0	15.1	10.9	8.2	(29)
(30)			MCDWB	10.5	11.5	14.7	19.9	23.8	26.4	27.2	27.1	23.6	18.9	13.4	10.4	(30)
(31)		5%	WB	6.1	6.7	9.3	12.3	16.1	19.1	19.5	19.3	17.1	14.1	9.7	6.7	(31)
(32)			MCDWB	7.9	9.4	13.1	17.9	21.9	25.2	26.1	26.1	21.6	17.2	11.6	8.3	(32)
(33)		10%	WB	4.6	5.3	8.2	11.3	14.9	18.1	18.7	18.5	16.1	13.0	8.3	5.3	(33)
(34)			MCDWB	6.0	7.3	11.6	16.4	19.9	23.6	24.6	24.5	19.8	15.5	9.8	6.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	7.6	9.0	10.5	10.3	10.4	10.8	10.9	10.1	8.8	6.2	5.6	(35)
(36)			MCDBR	8.6	11.3	14.2	15.8	14.9	14.6	15.2	15.4	14.4	12.9	10.4	8.5	(36)
(37)			MCWBR	6.3	7.6	8.6	8.3	7.1	6.6	6.1	6.1	7.5	8.0	7.4	6.3	(37)
(38)		5% WB	MCDWR	8.0	10.4	12.6	13.9	13.3	13.0	13.2	13.7	12.6	11.7	9.7	7.8	(38)
(39)			MCWBR	6.1	7.2	7.9	7.8	6.7	6.2	5.7	6.0	6.9	7.6	7.1	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.325	0.359	0.393	0.402	0.406	0.411	0.405	0.380	0.351	0.323	0.297	(40)	
(41)		taud	2.431	2.371	2.327	2.260	2.269	2.293	2.274	2.311	2.355	2.431	2.466	2.457	(41)	
(42)		Ebn at Noon	772	828	852	855	860	857	846	834	819	787	737	730	(42)	
(43)		Edh at Noon	65	86	106	125	129	127	128	118	102	81	63	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.03	1.92	3.03	4.30	5.00	5.57	5.36	4.70	3.38	2.12	1.12	0.82	(44)	
(45)		RadStd	0.15	0.26	0.38	0.56	0.58	0.48	0.49	0.47	0.42	0.25	0.14	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	+0.36	+0.37	N/A	N/A	N/A	+20		
(47) Regional (83 neighbors)		+0.41	N/A	N/A	N/A	+0.32	+0.33	N/A	N/A	+62	+24		

Nomenclature: See separate page